

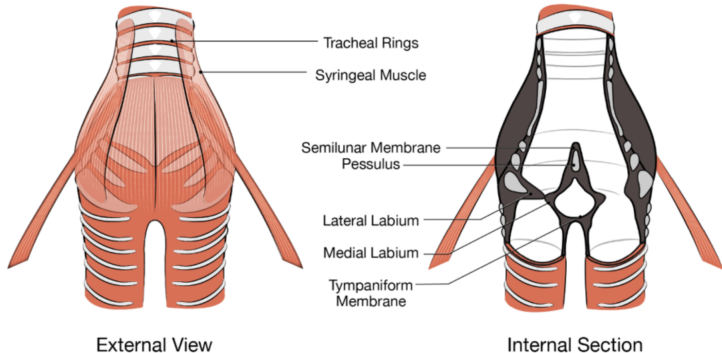


## Vocalizations

Dr. Ethan Linck  
Office: 305 Lewis Hall

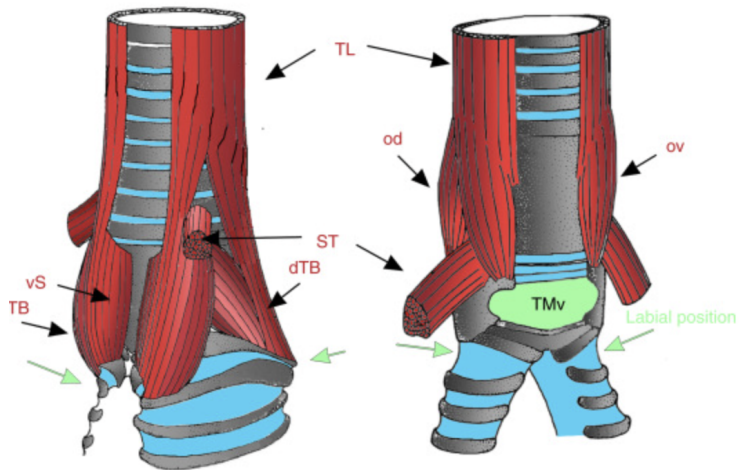
# The avian syrinx

- ▶ Located at junction of bronchi and trachea (c.f. **larynx** at top of neck)
- ▶ Sounds produced by forcing air through syrinx, reducing static pressure on syringeal membranes flanking interclavicular air sac (Bernoulli effect!)



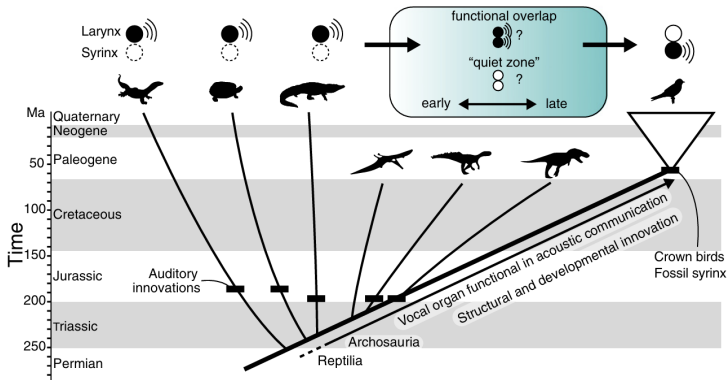
## Syrinx anatomy varies with vocal ability

- ▶ L: typical oscine songbird syrinx with both **tracheal** and **bronchial** elements
- ▶ R: suboscine song syrinx with only tracheal elements



# Syrinx evolution

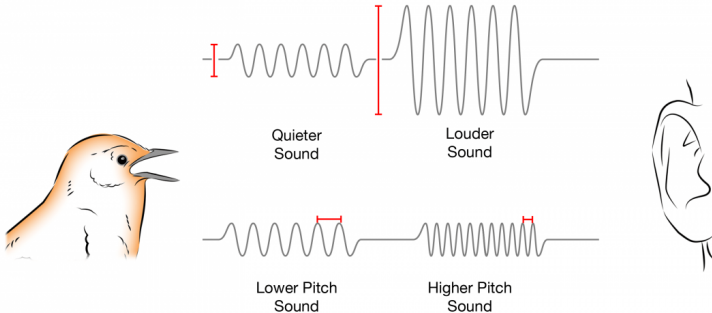
- ▶ ancestor of birds lost ability to make sounds with **larynx**, re-evolved ability with **syrinx** (an **evolutionary novelty**)



(Kinglsey et al. 2018)

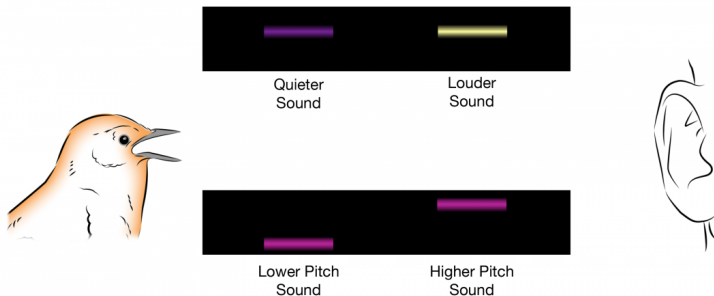
# Bird song physics

- ▶ **amplitude:** height of soundwaves = “loudness”
- ▶ **frequency:** distance between peaks of soundwaves = pitch



# Spectrograms

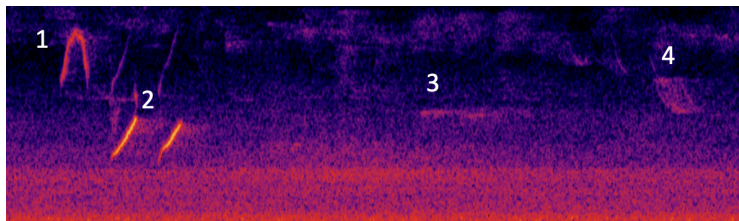
- **Spectrograms** visualize both amplitude and frequency



[https://macaulaylibrary.org/asset/61838;](https://macaulaylibrary.org/asset/61838)

<https://macaulaylibrary.org/asset/27193>

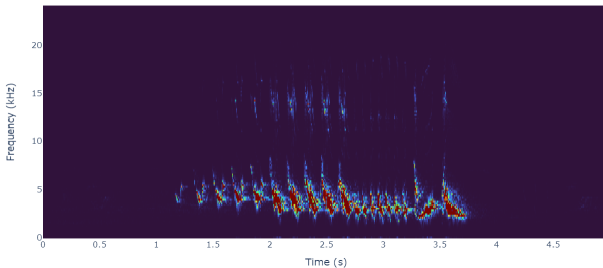
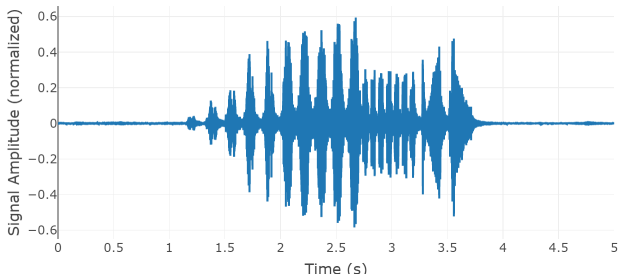
Which feature of this song is *loudest*?



<https://xeno-canto.org/1183619>

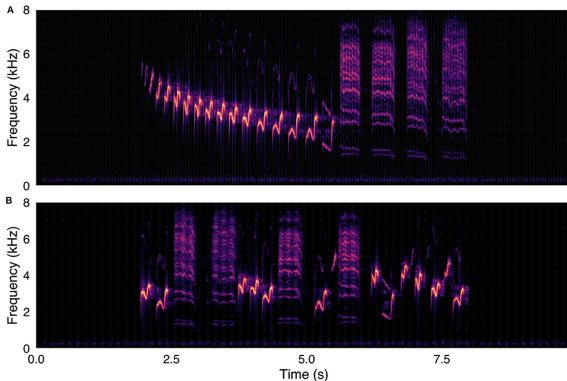
- a) 1
- b) 2
- c) 3
- d) 4

# Data from spectrograms and waveforms used in Merlin



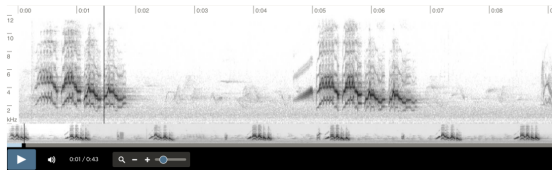
# Harmonics

- ▶ Multiple sound producing areas in syrinx = multiple simultaneous notes
- ▶ Harmonics are multiples of a “fundamental frequency”



<https://macaulaylibrary.org/asset/224190551>

# Self-duetting

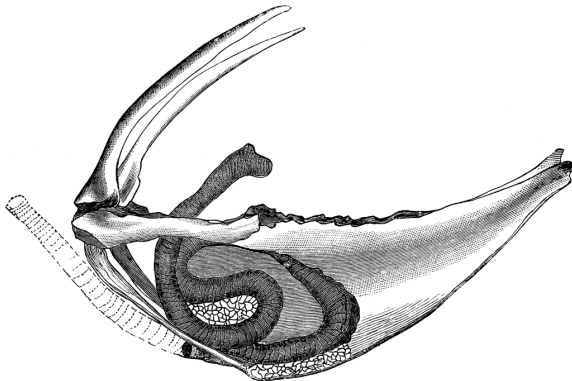


<https://earbirding.com/blog/archives/3711>

ML476610461; ML3093086;

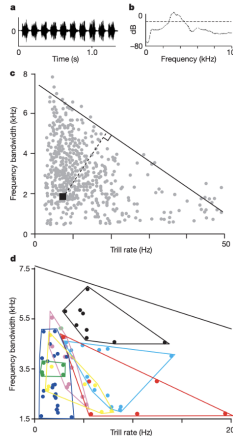
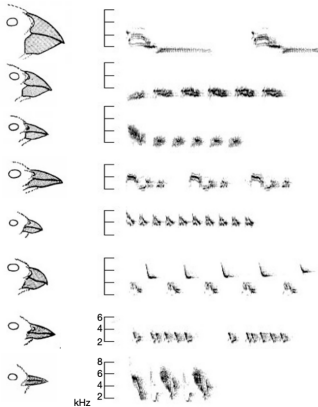
## Post-source modulation

- ▶ **filtering** eliminates or enhances certain vocal frequencies by interference with features in vocal cavity (e.g., bill)
- ▶ **post-source modulation** refers to altering sounds from syrinx via changes in length, shape, size of vocal cavity



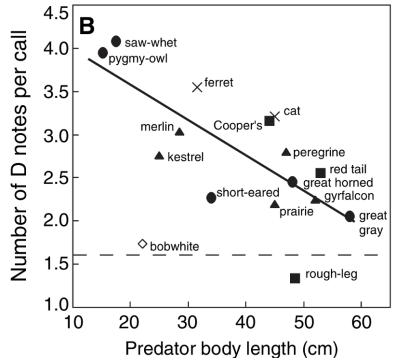
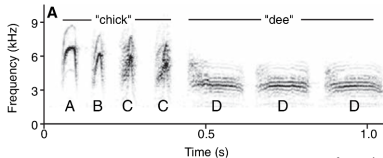
# Feeding ecology constrains song

- ▶ Bill size reflects adaptations to seed size
- ▶ High rate, pure-tone trills are physically impossible for birds with large bills



# Chickadees encode information on predator size in song

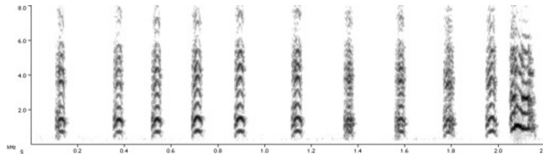
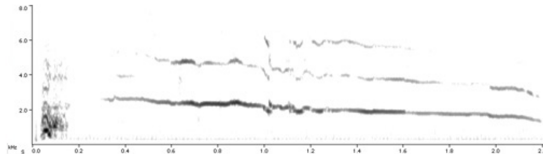
- ▶ Number of harmonic “dees” negatively correlates to predator body size (smaller = bigger threat)



(Podos 2001)

# Binaural comparison to locate predators

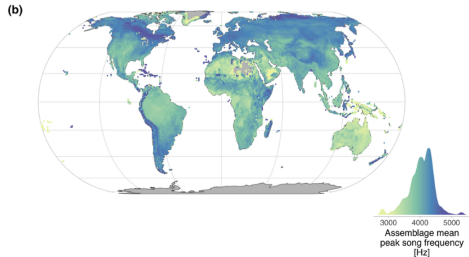
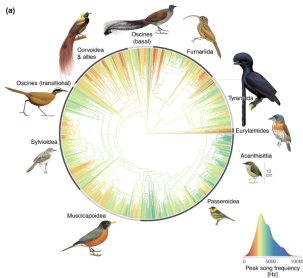
- ▶ **aerial predators** (top): limited amplitude, long waves scatter, difficult to source
- ▶ **ground predators** (bottom): large amplitude, short, lots of location information



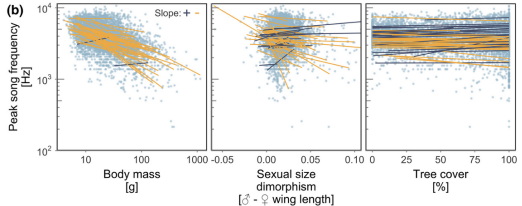
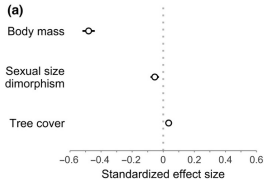
(*Gallus gallus* calls, Aves Biology / Gary Ritchison)

# Acoustic adaptation hypothesis

- ▶  $H_1$ : Birds in denser habitats will have lower calls
- ▶ Methods: Meta-analysis of frequencies paired with habitat cover



# Is the acoustic adaptation hypothesis supported?



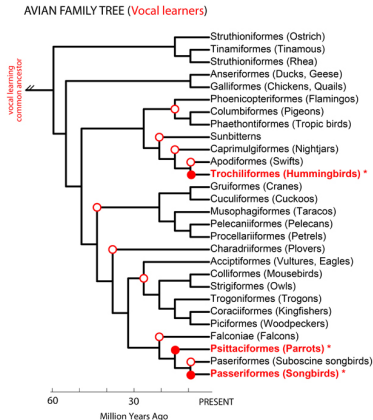
- a) yes
- b) no
- c) impossible to say
- d) it depends

(But see: <https://www.youtube.com/watch?v=iy-9Z2KrjsY>)

(Mikula et al. 2021)

# Vocal learning is rare but has evolved multiple times

- Found in parrots, songbirds, hummingbirds, humans, bats, and whales

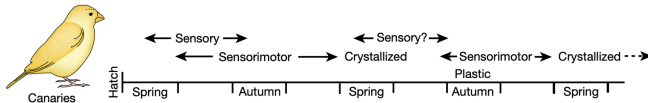
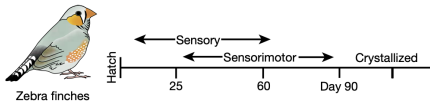
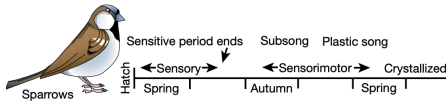


\* Vocal learners: Alternative hypotheses

- 3 independent gain
- 9 independent losses
- every animal has it to varying degrees

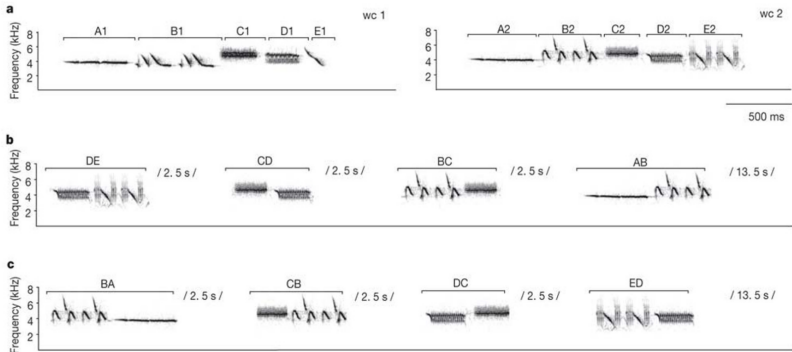
# Phases of song learning

- ▶ **sensory acquisition phase:** innate “template” modified by auditory environment
- ▶ **sensorimotor phase:** practices **subsong** to develop **plastic** (flexible but complete) song
- ▶ **crystalized phase:** stable adult song



# Phases of song learning

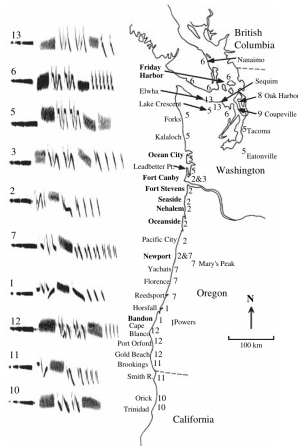
- White-crowned Sparrows only use phrases from “tutors”



(Rose et al. 2004)

# Learned song variation

- ▶ **dialects:** geographically constrained song variants
- ▶ **repertoire:** the set of songs an individual knows



(Nelson et al. 2004)

Which of the following will influence calls in tinamous?



- a) Bill size
- b) Habitat
- c) Parental repertoire
- d) Syrinx anatomy

*Tinamotis ingoufi*

# Mimicry

- ▶ ~20% of passerine songbirds practice mimicry



<https://www.youtube.com/watch?v=mSB71jNq-yQ>

## References

- ▶ Title slide species: *Anthreptes anchietae*, *Rhipidura versicolor*, *Mimus thenca*, *Ortalis wagleri*
- ▶ Goller, F. (2022). The syrinx. *Current Biology*, 32(20), R1095-R1100.
- ▶ Kingsley, E. P., Eliason, C. M., Riede, T., Li, Z., Hiscock, T. W., Farnsworth, M., ... & Clarke, J. A. (2018). Identity and novelty in the avian syrinx. *Proceedings of the National Academy of Sciences*, 115(41), 10209-10217.
- ▶ Bilger, H. T., Vertosick, E., Vickers, A., Kaczmarek, K., & Prum, R. O. (2021). Higher-order musical temporal structure in bird song. *Frontiers in Psychology*, 12, 629456.
- ▶ Coues, E. (1882). The Coues check list of North American birds. Estes and Lauriat.

## References

- ▶ Podos, J. (2001). Correlated evolution of morphology and vocal signal structure in Darwin's finches. *Nature*, 409(6817), 185-188.
- ▶ Templeton, C. N., Greene, E., & Davis, K. (2005). Allometry of alarm calls: black-capped chickadees encode information about predator size. *Science*, 308(5730), 1934-1937.
- ▶ Mikula, P., Valcu, M., Brumm, H., Bulla, M., Forstmeier, W., Petrusková, T., ... & Albrecht, T. (2021). A global analysis of song frequency in passerines provides no support for the acoustic adaptation hypothesis but suggests a role for sexual selection. *Ecology Letters*, 24(3), 477-486.
- ▶ Petkov, C. I., & Jarvis, E. D. (2012). Birds, primates, and spoken language origins: behavioral phenotypes and neurobiological substrates. *Frontiers in evolutionary neuroscience*, 4, 12.

## References

- ▶ Brainard, M. S., & Doupe, A. J. (2002). What songbirds teach us about learning. *Nature*, 417(6886), 351-358.
- ▶ Rose, G. J., Goller, F., Gritton, H. J., Plamondon, S. L., Baugh, A. T., & Cooper, B. G. (2004). Species-typical songs in white-crowned sparrows tutored with only phrase pairs. *Nature*, 432(7018), 753-758.
- ▶ Nelson, D. A., Hallberg, K. I., & Soha, J. A. (2004). Cultural evolution of Puget sound white-crowned sparrow song dialects. *Ethology*, 110(11), 879-908.